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Integration Testing in Software Product Line Engineering: A Model-Based Technique

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Fundamental Approaches to Software Engineering

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Abstract

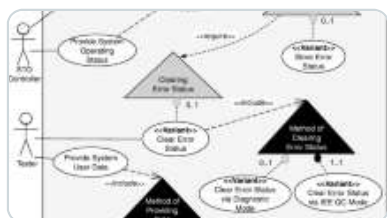
The development process in software product line engineering is divided into domain engineering and application engineering. As a consequence of this division, tests should be performed in both processes. However, existing testing techniques for single systems cannot be applied during domain engineering, because of the variability in the domain

artifacts. Existing software product line test techniques only cover unit and system tests. Our contribution is a model-based, automated integration test technique that can be applied during domain engineering. For generating integration test case scenarios, the technique abstracts from variability and assumes that placeholders are created for variability. The generated scenarios cover all interactions between the integrated components, which are specified in a test model. Additionally, the technique reduces the effort for creating placeholders by minimizing the number of placeholders needed to execute the integration test case scenarios. We have experimentally measured the performance of the technique and the potential reduction of placeholders.

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Keywords

[Test Model](#)

[Variation Point](#)

[Software Product Line](#)

[Variable Part](#)

[Domain Engineering](#)

These keywords were added by machine and not by the authors. This process is experimental and the keywords may be updated as the learning algorithm improves.

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